HOMOMORPHISMS OUT OF A FINITELY PRESENTED MODULE COMMUTE WITH FILTERED COLIMITS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The compatibility of filtered colimits of modules with restriction to open subsets. Invoked at the start of the counted proof (source lines 3990-3991) as a cross-chapter pointer. Reduces the statement to the case of a module admitting a global presentation.
- The exactness of the sheaf of homomorphisms in the first variable for a right exact sequence. Invoked inside the counted proof at source line 4006 as a pointer to a lemma of this chapter. Produces the two exact rows of the diagram whose outer vertical arrows are computed directly.

Source reference. Original lemma lemma-internal-hom-exact: sources/stacks-modules/pinned-original-source.tex, source lines 3887–3887. Navigation pointer only: the counted text refers to the Stacks item lemma-internal-hom-exact, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference sheaves-section-limits-sheaves: sources/stacks-modules/pinned-original-source.tex, source lines 2640–2640. Navigation pointer only: the counted text refers to the Stacks item sheaves-section-limits-sheaves, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Date: stacks-project:modules.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let X be a ringed space. Let $\mathcal{F}$ be an $\mathcal{O}_X$-module of finite presentation. Let $\mathcal{G} = \operatorname{colim}_{\lambda \in \Lambda} \mathcal{G}_\lambda$ be a filtered colimit of $\mathcal{O}_X$-modules. Then the canonical map*

$$\operatorname{colim}_\lambda \mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G}_\lambda) \longrightarrow \mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})$$

is an isomorphism.

Proof. Taking colimits of sheaves of modules commutes with restriction to opens, see Sheaves, Section sheaves-section-limits-sheaves. Hence we may assume $\mathcal{F}$ has a global presentation

$$\bigoplus\nolimits_{j=1,\ldots,m} \mathcal{O}_X \longrightarrow \bigoplus\nolimits_{i=1,\ldots,n} \mathcal{O}_X \to \mathcal{F} \to 0$$

The functor $\mathcal{H}om_{\mathcal{O}_X}(-,-)$ commutes with finite direct sums in either variable and $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{O}_X, -)$ is the identity functor. By this and by Lemma lemma-internal-hom-exact we obtain an exact sequence

$$0 \to \mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G}) \to \bigoplus\nolimits_{i=1,\ldots,n} \mathcal{G} \to \bigoplus\nolimits_{j=1,\ldots,m} \mathcal{G}$$

Since filtered colimits are exact in $Mod(\mathcal{O}_X)$ also the top row in the following commutative diagram is exact

$$\begin{array}{ccccccc}
0 & \longrightarrow & \operatorname{colim}_\lambda \mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G}_\lambda) & \longrightarrow & \operatorname{colim}_\lambda \bigoplus_{i=1,\ldots,n} \mathcal{G}_\lambda & \longrightarrow & \operatorname{colim}_\lambda \bigoplus_{j=1,\ldots,m} \mathcal{G}_\lambda \\
 & & \downarrow & & \downarrow & & \downarrow \\
0 & \longrightarrow & \mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G}) & \longrightarrow & \bigoplus_{i=1,\ldots,n} \mathcal{G} & \longrightarrow & \bigoplus_{j=1,\ldots,m} \mathcal{G}
\end{array}$$

Since the right two vertical arrows are isomorphisms we conclude. □